

Progetto Neural Network

Neapolitan Cards Recognition

Autori

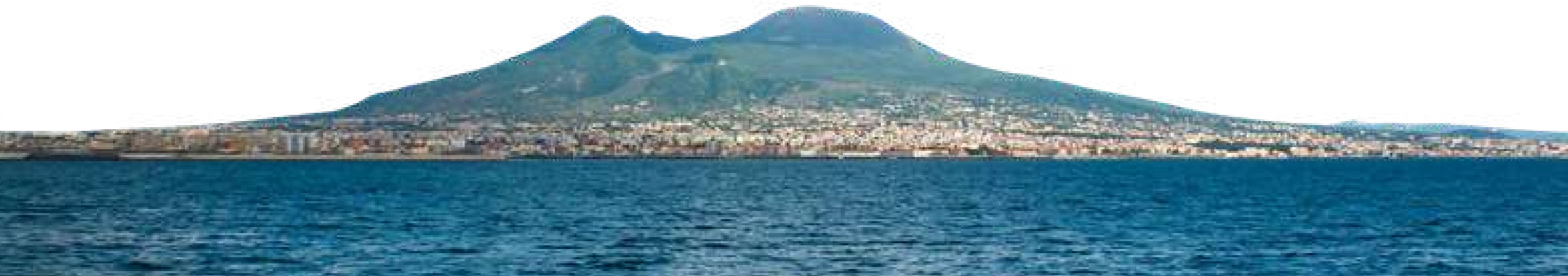
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Motivazione

Esistono molti esempi e dataset pubblici
per carte francesi

Nessuna soluzione per le carte napoletane



Dataset

Sperimentazione 1

- 3 mazzi nel train set (3 modiano)
- 1 mazzi nel validation set (1 modiano)
- 1 mazzi nel test set (1 del negro)



Sperimentazione finale

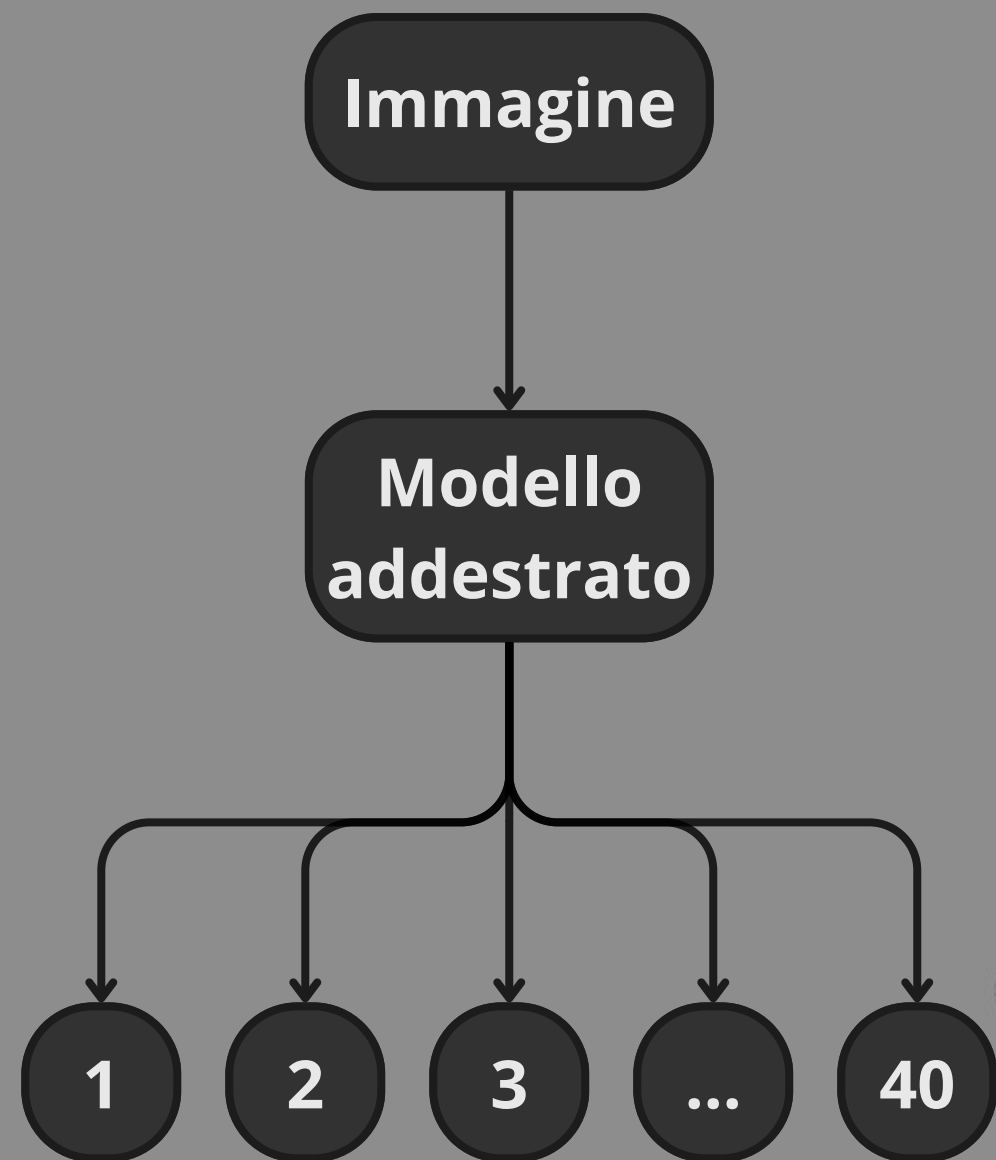
- 6 mazzi nel train set (6 modiano)
- 2 mazzi nel validation set (2 modiano)
- 4 mazzi nel test set (3 modiano, 1 del negro)



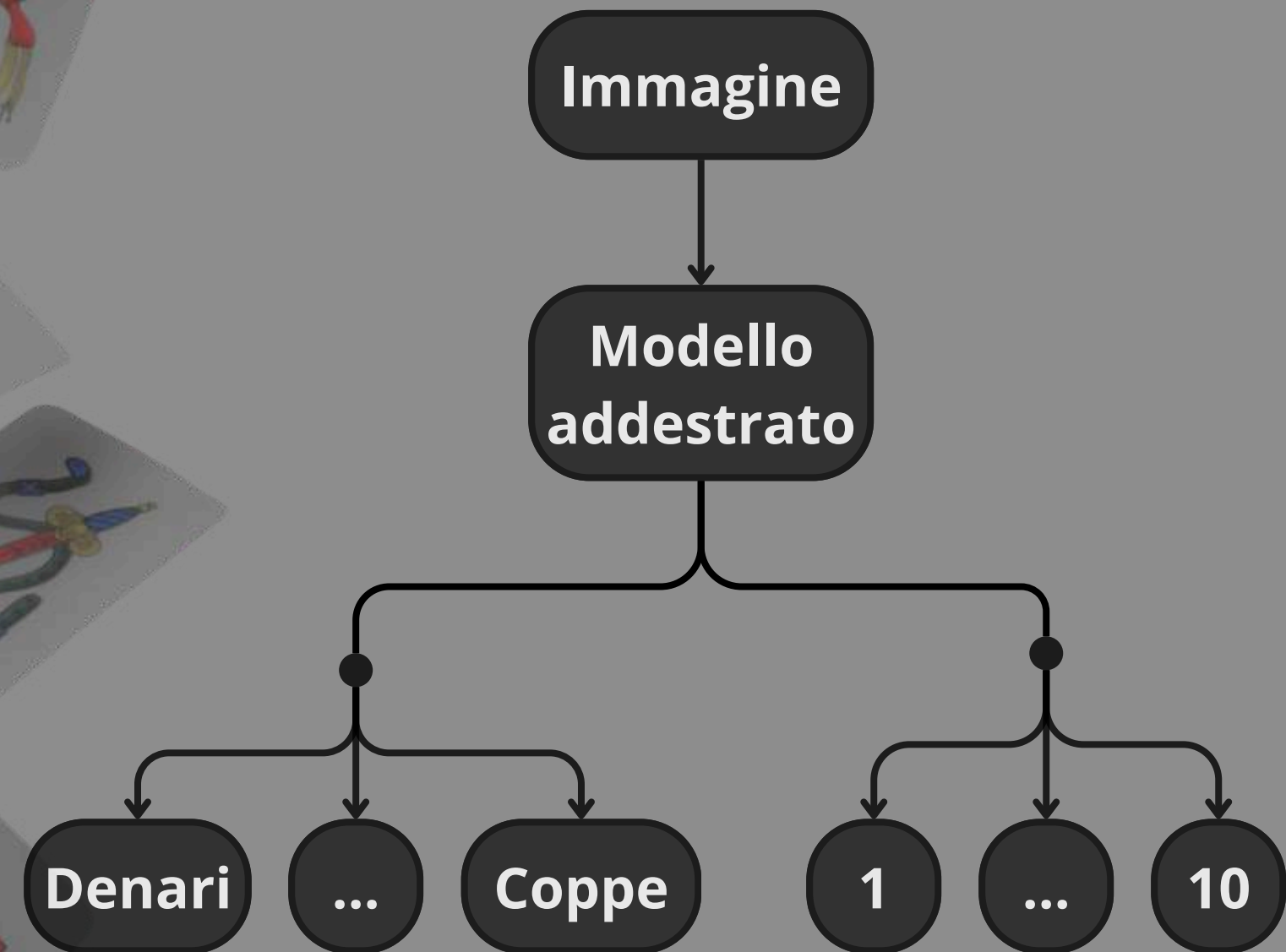
[Public Roboflow](#)

Approcci utilizzati

Modello a 40 classi



Modello multi-head



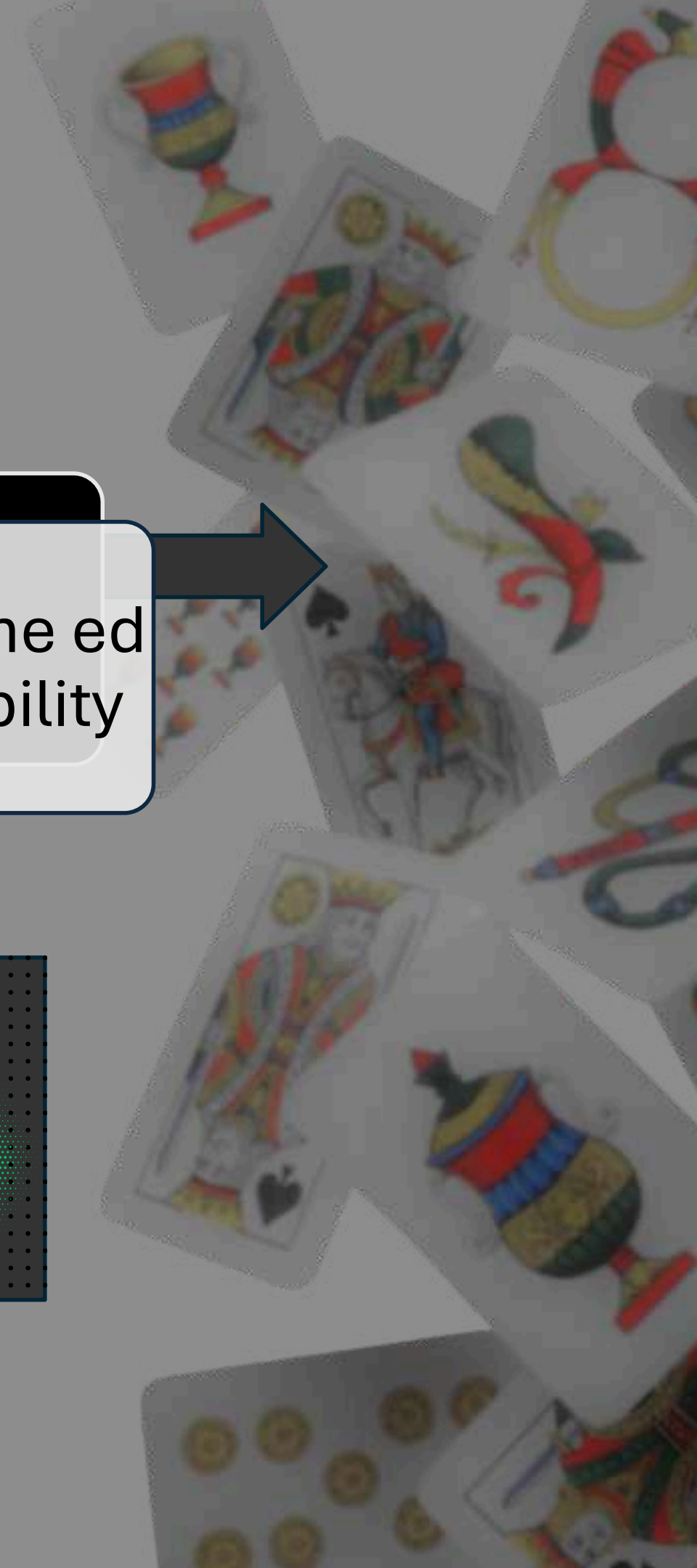
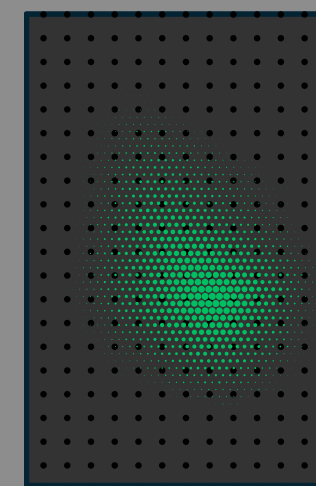
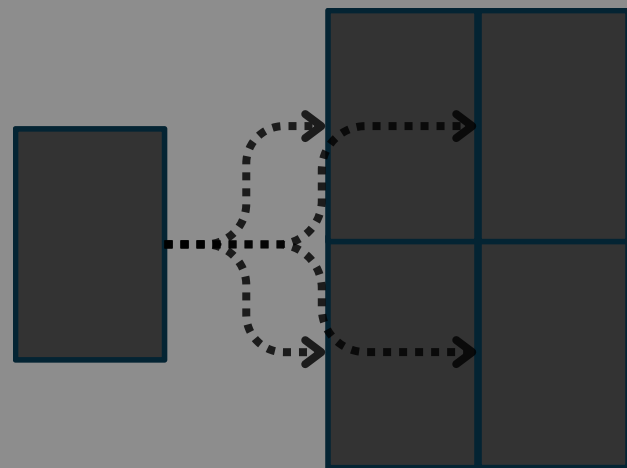
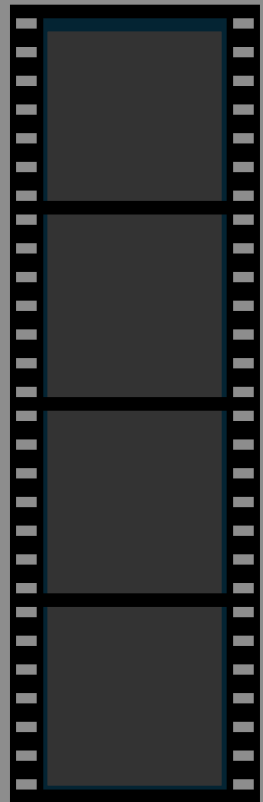
Deep learning pipeline

Acquisizione
dataset

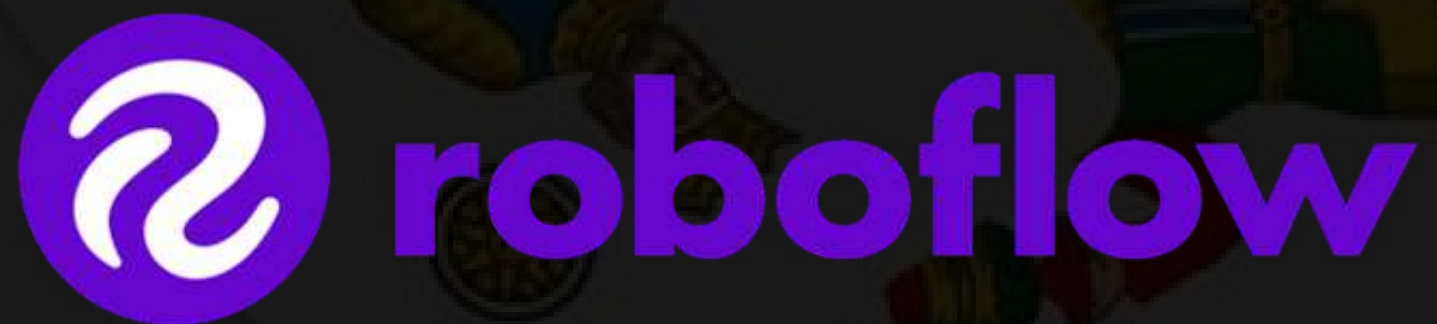
Preprocessing e
augmentation

Addestramento

Valutazione ed
explainability



Tecnologie utilizzate



Risultati ottenuti



[Link ai risultati visualizzabili](#)

The background of the image is a collage of various Tarot cards. Visible cards include the King of Swords (top left), the King of Wands (top right), the Knight of Wands (bottom right), and the Knight of Cups (bottom left). There are also several smaller cards showing symbols like swords, wands, and chalices. A dark, semi-transparent rectangular box is centered over the image, containing the text.

Ma funziona
davvero?

Img 147 | Real: 9 A | Pred: 9 A

Originale



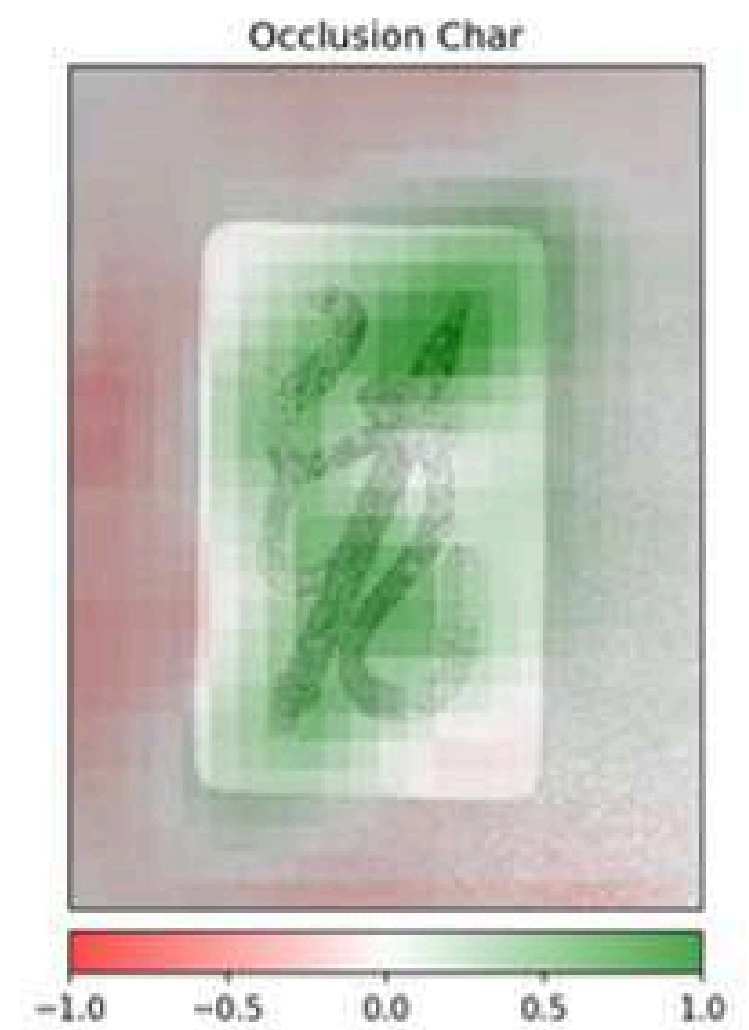
IG SmoothGrad



Occlusion



Img 9 | Real: 1-C | Pred: 1-C





Generalizza?

Img 67 | Real: 4 A | Pred: 4 A

Originale



IG SmoothGrad



0.0 0.2 0.4 0.6 0.8 1.0

Occlusion



-1.0 -0.5 0.0 0.5 1.0

Img 66 | Real: 4 A | Pred: 4 A

Originale

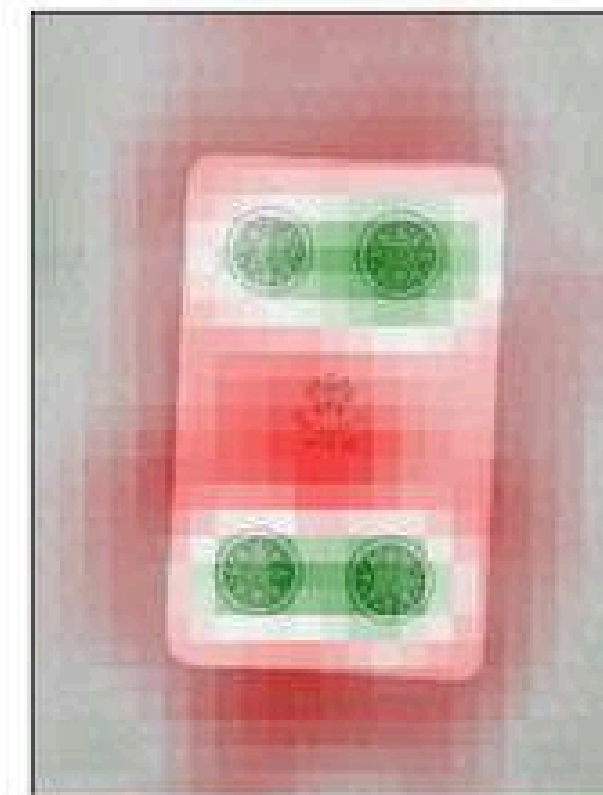


IG SmoothGrad



0.0 0.2 0.4 0.6 0.8 1.0

Occlusion

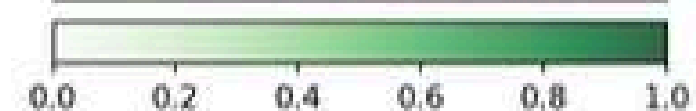


-1.0 -0.5 0.0 0.5 1.0

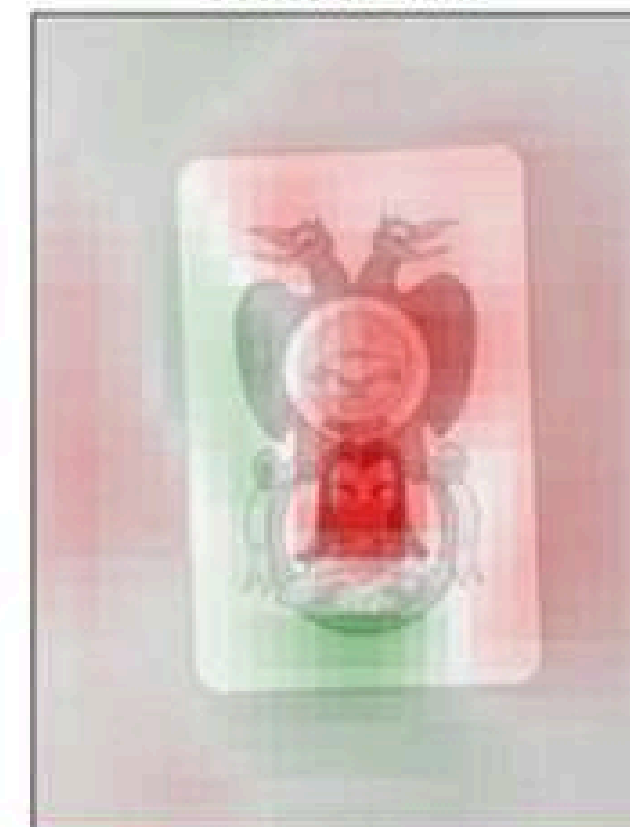
Orig (Pred: 1 | 47.5%)



IG Smooth Num



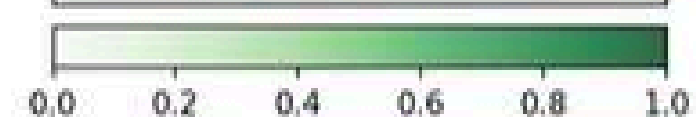
Occlusion Num



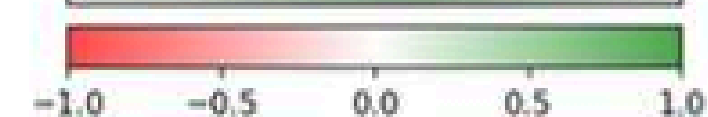
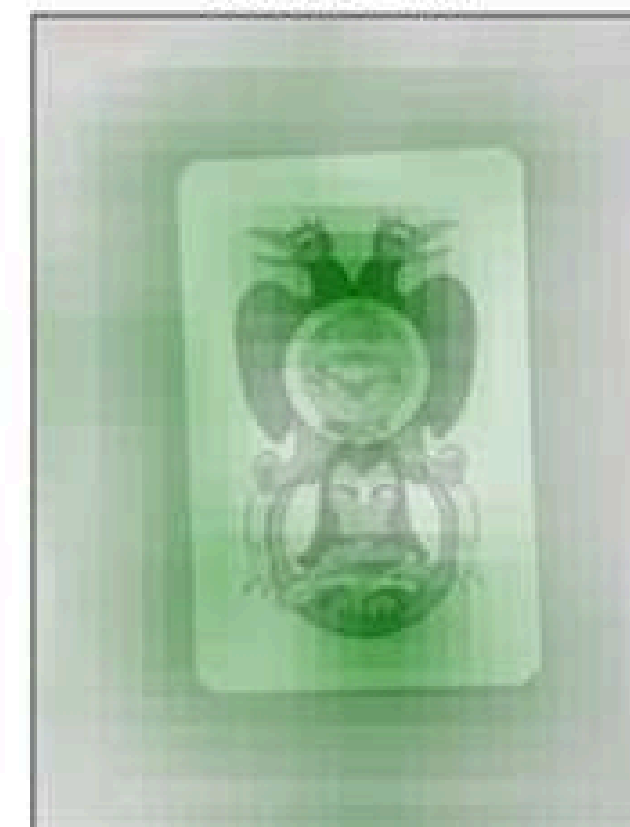
Orig (Pred: A | 99.5%)



IG Smooth Char



Occlusion Char



Grazie per l'attenzione

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[GitHub](#)